AP PRECALCULUS

Unit 1 • Worksheet 1.1-1.6A • Answer key

Review 1.1-1.6A • Review of Topics 1.1 to 1.6

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1 (1.2)

NO CALCULATOR

What is the average rate of change of $f(x) = x^3 - x$ on $[0, 2]$?

(A) 6
(B) 3
(C) 2
(D) 4

Solution

Answer: (B). $f(2) = 6$, $f(0) = 0$: $\frac{6}{2} = 3$.

Question 2 (1.3)

NO CALCULATOR

Outputs at $x = 0, 1, 2, 3, 4$ are $1, 4, 9, 16, 25$. Which model fits?

(A) linear
(B) quadratic
(C) cubic
(D) none

Solution

Answer: (B). Second differences 2.

Question 3 (1.5)

NO CALCULATOR

A polynomial with real coefficients has degree 3 and zeros 4 and $-i$. What is the third zero?

(A) -4
(B) i
(C) $4i$
(D) 0

Solution

Answer: (B). Conjugate of $-i$.

Question 4 (1.6) NO CALCULATOR

Which is $\lim_{x \to -\infty} \left(3x^4 - x^5\right)$?

(A) $-\infty$
(B) ∞
(C) 3
(D) 0

Solution
Answer: (B). Leading term $-x^5$; as $x \to -\infty$, $-x^5 \to \infty$.

Question 5 (1.4) NO CALCULATOR

How many turning points can a polynomial of degree 4 have?

(A) only 3
(B) 1 or 3
(C) 0, 1, 2 or 3
(D) 4

Solution
Answer: (B). A degree-4 polynomial has at least one and at most three turning points, and the number is odd (end behaviors match). So 1 or 3.

Question 6 (Free response) NO CALCULATOR

Let $p(x) = x^3 - 3x^2 - x + 3$.

(a) Factor p completely and find its zeros (hint: group).
(b) Find the average rate of change of p on $[-1, 1]$.
(c) Describe the end behavior and sketch the graph.

Solution
(a) $x^2(x-3) - (x-3) = (x-3)(x-1)(x+1)$: zeros $3, 1, -1$.
(b) $p(1) = 0$, $p(-1) = 0$: average rate of change 0.
(c) $\lim_{x \to \infty} p = \infty$, $\lim_{x \to -\infty} p = -\infty$. Rises through -1, peaks above $(0, 3)$, falls through 1, turns below the axis, rises through 3.